

To: Head of Product, Scout

From: Edward Ge

Subject: Product Prioritization for Recommendations and Price-Drop Alerts

Date: July 2026

Business Question

Which products should Scout prioritize for personalized recommendations and price-drop alerts?

Recommendation

Scout should separate product prioritization into two tracks: repeated-demand products for recommendations and higher-ticket products for price-drop alerts. For recommendations, Scout should prioritize products with strong revenue and consistent order activity, such as **WHITE HANGING HEART T-LIGHT HOLDER, JUMBO BAG RED RETROSPOT, PARTY BUNTING, ASSORTED COLOUR BIRD ORNAMENT, and REGENCY CAKESTAND 3 TIER**. For price-drop alerts, Scout should test higher-priced home and furniture-style products such as **RUSTIC SEVENTEEN DRAWER SIDEBBOARD, VINTAGE BLUE KITCHEN CABINET, VINTAGE RED KITCHEN CABINET, LOVE SEAT ANTIQUE WHITE METAL, and SET/4 WHITE RETRO STORAGE CUBES**, where discounts may be more noticeable to shoppers.

Key Evidence

I used the UCI Online Retail dataset and cleaned the data by removing cancelled invoices, invalid quantities, non-positive prices, and non-product records such as postage, Amazon fees, bank charges, manual adjustments, and bad debt adjustments. The analysis also showed why revenue alone is not enough: PAPER CRAFT, LITTLE BIRDIE generated very high revenue but appeared in only one order, suggesting a one-off bulk purchase. To avoid this issue, I used a weighted recommendation score based on revenue rank, order count rank, and quantity rank, while selecting price-drop candidates with a separate high-ticket logic.

Limitations and Next Steps

This dataset does not include browsing behavior, saved items, wishlist activity, or actual price-change history. As a next step, Scout should validate these product priorities using internal user metrics such as saves, clicks, alert opens, and conversions after price-drop notifications.